

$$\begin{aligned}
& (\neg P \vee \neg Q) \rightarrow \neg(P \leftrightarrow \neg Q) \\
& \equiv (\neg P \vee \neg Q) \rightarrow ((P \rightarrow \neg Q) \wedge (\neg Q \rightarrow P)) \text{ (Section 1.3, Table 8, Equivalence 1)} \\
& \equiv (\neg P \vee \neg Q) \rightarrow ((\neg P \vee \neg Q) \wedge (\neg(\neg Q) \vee P)) \text{ (Conditional Disjunction Equivalence)} \\
& \equiv (\neg P \vee \neg Q) \rightarrow ((\neg P \vee \neg Q) \wedge (Q \vee P)) \text{ (Double-Negation Law)} \\
& \equiv \neg(\neg P \vee \neg Q) \vee ((\neg P \vee \neg Q) \wedge (Q \vee P)) \text{ (Conditional Disjunction Equivalence)} \\
& \equiv \neg(\neg(P \wedge Q)) \vee ((\neg P \vee \neg Q) \wedge (Q \vee P)) \text{ (1st De Morgan's Law)} \\
& \equiv (P \wedge Q) \vee ((\neg P \vee \neg Q) \wedge (Q \vee P)) \text{ (Double Negation Law)} \\
& \equiv ((P \wedge Q) \vee (\neg P \vee \neg Q)) \wedge ((P \wedge Q) \vee (Q \vee P)) \text{ (1st Associative Law)} \\
& \equiv ((P \wedge Q) \vee \neg(P \wedge Q)) \wedge ((P \wedge Q) \vee (Q \vee P)) \text{ (1st De Morgan's Law)} \\
& \equiv T \wedge ((P \wedge Q) \vee (Q \vee P)) \text{ (1st Negation Law)} \\
& \equiv ((P \wedge Q) \vee (Q \vee P)) \wedge T \text{ (2nd Commutative Law)} \\
& \equiv ((P \wedge Q) \vee (Q \vee P)) \text{ (1st Identity Law)} \\
& \equiv ((Q \vee P) \vee (P \wedge Q)) \text{ (1st Commutative Law)} \\
& \equiv (Q \vee (P \vee (P \wedge Q))) \text{ (1st Associative Law)} \\
& \equiv (Q \vee P) \text{ (1st Absorption Law)} \\
& \equiv (P \vee Q) \text{ (1st Commutative Law)}
\end{aligned}$$

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